TALENTQ

# The Technical Blueprint

Evidence-driven matching. Two-sided compatibility.
Governed AI recruitment from discovery to hiring.

PROPOSED SYSTEM • V1

Architecture / mathematical models / data governance /
latency / agent workflows / validation / build gates

25 OCTOBER 2026

STRATEGY / ENGINEERING / INVESTOR DILIGENCE

TECHNICAL SERIES / DOCUMENT 01

# Document purpose

A proposed technical specification for a standalone AI-native recruitment platform. It is intended to guide architecture, experimentation, product requirements and future investor diligence. It is **not** evidence of a deployed, validated or patented matching system.

> **Independent product narrative**
> TalentQ stands on its own merits. The external product identity, mission, performance claims and investment narrative should not depend on another application. Internally, engineering teams may reuse existing governed Q capabilities where security, performance and modularity tests justify doing so.

## Decisions this document makes

- Defines a candidate-evidence system with provenance, freshness and candidate permissions.
- Separates job eligibility, employer competency fit and candidate preference fit.
- Specifies a transparent Q-Match baseline with missing-data treatment, uncertainty and ranking diagnostics.
- Defines agent scopes, interview orchestration, measurable speed budgets and reliability controls.
- Sets validation gates before claims about predictive accuracy, fairness, autonomous operation or defensibility.

## Decisions it deliberately does not make

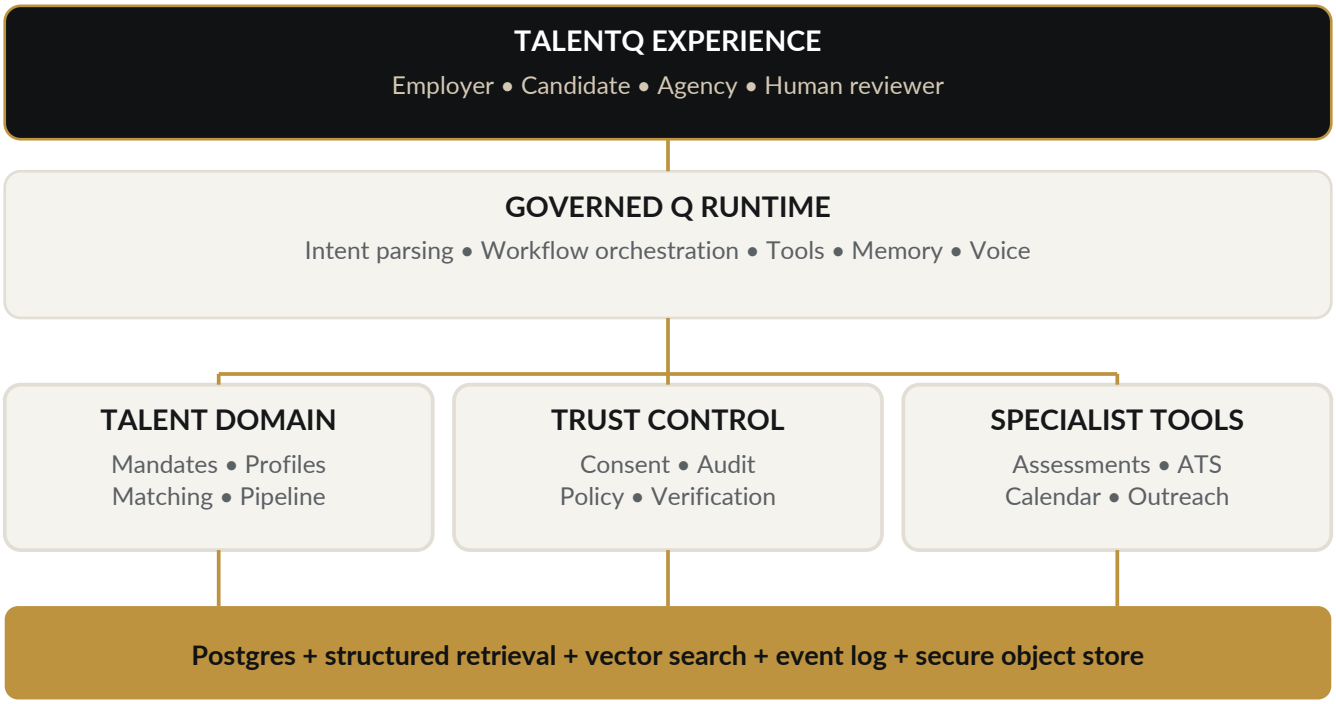
- Market size, realistic pricing, competitor economics, country-by-country willingness to pay or an investor return multiple. These require the dedicated market report.
- Legal approval for any jurisdiction; the technical controls here require qualified legal review.
- A claim that a mathematical equation or a model is patentable, unique or already trained.
- Any promise of unrestricted LinkedIn data access or instant completion of every external search.

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**Notation:** F = employer-side competency fit; A = candidate-side compatibility; M = mutual-fit index; C = evidence coverage; U = uncertainty; S = optimized shortlist. Unless explicitly labeled as sourced facts, all thresholds, examples and architecture choices are proposed design parameters.

DESIGN PRINCIPLES

# Executive system summary




TalentQ has four users: employer, agency recruiter, candidate and human reviewer. Each interacts with one governed conversational system backed by deterministic recruiting services and traceable evidence.

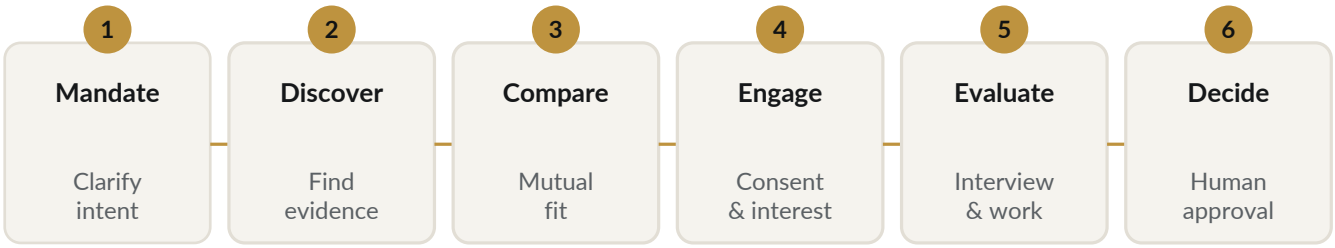
> **North-star technical principle**
> An LLM understands requirements and explains evidence. Deterministic services validate permissions, calculate scores, record events and enforce transitions. AI agents may propose or execute authorized tasks, but cannot silently become the final hiring authority.

## Seven architectural invariants

- No ranking without a defined role, versioned rubric and source-traceable evidence.
- No cross-tenant sharing without explicit access rights; candidate-owned information stays permission-scoped.
- No surprise candidate outreach, background investigation or reuse of private practice recordings.
- Unknown, stale and contradicted evidence remain visibly uncertain.
- Human review remains meaningful for consequential selection decisions.
- All important outputs are reproducible from their data, model, policy and rubric versions.
- Every screen responds promptly even when external discovery or verification is still working.

01 / END-TO-END WORKFLOW

# The recruitment journey as a state machine




## State transitions

| Object | Valid states | Transition authority |
|---|---|---|
| Hiring mandate | draft → clarified → approved → sourcing → review → closed | Employer or authorized recruiter; Q may draft |
| Candidate representation | discovered → eligible-to-contact → invited → engaged → withdrawn | Policy engine + candidate choice |
| Application | interested → screened → assessed → interview → decision → offer → hired | Authorized recruiter or hiring manager |
| Assessment | assigned → consented → started → submitted → reviewed → expired | Candidate, specialist provider, reviewer |
| Outreach | drafted → approved → sent → replied / opted-out | User approval or tightly scoped standing delegation |

## Where automation stops

Q can generate search strategies, candidate comparisons, invitation drafts, assessment recommendations, rescheduling suggestions and status summaries. A human owns job approval, assessment exceptions, final advancement or rejection decisions, offers and material changes to applicant rights.

**Failure-first requirement**
If external data access fails, the workflow enters **source unavailable**, not **no candidates exist**. If a candidate preference is unknown, the system asks or reports uncertainty, not an invented willingness to relocate or accept pay.

02 / SERVICE ARCHITECTURE

# Modules and reusable contracts

| Module | Responsibility | Primary outputs |
|---|---|---|
| M01 Mandate compiler | Convert hiring conversation into approved requirements | Versioned job rubric + eligibility rules |
| M02 Candidate identity & consent | Profiles, roles, preferences, visibility, opt-outs | Authorized identity and grants |
| M03 Evidence graph | Normalize claims, skill evidence, verification and freshness | Traceable competency evidence |
| M04 Source connectors | Ingest permitted external and employer-owned records | Source events + sync diagnostics |
| M05 Discovery & matching | Hybrid search, eligibility, fit and slate ranking | Ranked provisional candidate set |
| M06 Engagement | Outreach, interest confirmation, messaging, scheduling | Communication timeline |
| M07 Interview & assessment | Voice/text interview and practical provider routing | Structured competency observations |
| M08 Agent runtime | Long-running governed sourcing and coordination | Scoped job records and action proposals |
| M09 Candidate growth | Private practice, feedback, credential updates | Candidate-controlled improvement plan |
| M10 Analytics & governance | Audit, fairness evaluation, A/B tests, monitoring | Versioned quality and risk metrics |

Services expose versioned tools and events. They must not depend on a specific assistant prompt, UI framework or voice provider. The orchestration layer can be reused internally, but TalentQ preserves separate branding, tenancy, policies and application data.

### Recommended core implementation

An API and workflow service with PostgreSQL for transactional entities, full-text indexes for exact attributes, a vector index for semantic retrieval, object storage for permitted artifacts, and an event/outbox pattern for reliable asynchronous work. Begin with a modular monolith and an isolated job-worker tier. Extract services only after operational evidence justifies the complexity.

03 / CANDIDATE DATA MODEL

# Evidence is the product

A profile is not a single AI-generated summary. It is a graph of claims with provenance, rights and temporal context. A generated narrative is only a view of that graph.

| Entity | Required fields | Rule |
|---|---|---|
| Candidate | candidate_id, tenant relationships, visibility policy | Candidate identity separate from organization record |
| Claim | subject, predicate, value, asserted_by, asserted_at | May remain unverified |
| Evidence | claim_id, source URI, captured_at, method, license, consent scope | Must be traceable |
| Verification | evidence_id, status, reviewer/provider, expiry | Credential checks never implied |
| Competency | ontology_id, role relevance, level rubric | Versioned skill taxonomy |
| Preference | candidate_id, field, value, last_confirmed | Candidate-editable, not inferred from stereotypes |
| Sharing grant | purpose, recipient/tenant, data scope, expiry, revocation | Enforced on every read and outbound action |

## Evidence reliability

$$r(e) = v(e) \times f(e) \times d(e)$$

v = validation quality, f = freshness discount, d = evidence relevance; all in [0,1]. Proposed heuristic, not a population-calibrated probability.

Freshness may use an exponential decay where appropriate: **f(e) = exp[-ln(2) × age / half-life]**. The half-life depends on evidence type. A work sample may be relevant for longer than a phone number or job-availability declaration.

### Conflict handling

- If two sources disagree about employment dates, show the conflicting claims and request clarification.
- Never silently rewrite third-party facts from an LLM inference.
- For de-duplication, require positive identity evidence; reversible merge operations are mandatory.
- Candidate edits do not falsify external records; they create correction events and review flags.

04 / DATA ACCESS

# Authorized acquisition and source strategy

The fastest way to ruin the data strategy is to treat a professional-network subscription as an unrestricted data API. LinkedIn Recruiter System Connect access requires approved partner status and data agreements [S1].

| Acquisition tier | Examples | Permitted use condition |
|---|---|---|
| Tier A / Native | Candidate CV, work history, portfolio, assessments | Explicit notice, purpose and candidate controls |
| Tier B / Employer | ATS records, referrals, previous applicants | Employer authorization and role-based scopes |
| Tier C / Licensed | Recruitment data feeds, verification or skill providers | Specific agreement, retention and redistribution rules |
| Tier D / Public professional | Personal websites, open repositories, professional directories | Lawful basis, platform terms and documented necessity |
| Tier E / Disallowed by default | Unapproved scraping, private social media, brokered sensitive dossiers | Never ingest without separate lawful and policy approval |

## Connector acceptance checklist

- Document identity, authorization mechanism, permitted purposes and contractual data restrictions.

- Define rate limits, freshness SLAs, deletion propagation and pagination behavior.
- Test partial sync, stale cursor, retries, webhook signatures, duplicate events and account revocation.
- Store the source contract version for every imported record and block impermissible downstream use.

> **Explicit scope boundary**
> A recruiter asking Q to inspect Instagram posts does not by itself establish that such investigation is necessary or lawful. Off-role social behavior, inferred personality, relationship graphs and protected attributes are excluded from routine candidate scoring.

05 / LANGUAGE TO STRUCTURE

# Mandate compilation: from speech to rubric

The recruiter begins with natural language. Q asks targeted follow-ups until the mandate has enough structure for defensible comparison. The employer approves the final specification before scoring or outreach.

## Mandate fields

| Field group | Examples | Treatment |
|---|---|---|
| Job outcomes | Maintain APIs; reduce errors; produce monthly risk reports | Translate into observable competencies |
| Eligibility | Legally required license, work authorization, essential shifts | Approved explicit gates; unknown prompts verification |
| Competencies | Incident triage, SQL, enterprise account management | Weights linked to job analysis |
| Constraints | Budget range, location, work arrangement, start date | Mutual compatibility dimensions |
| Assessments | Interview dimensions, work sample, accessibility alternatives | Consistent versioned plans |
| Permissions | May Q contact? use ATS? share profiles? schedule? | Mandate-scoped action capabilities |

$$R = \{J, H, K, W, P, T, G, V\}$$

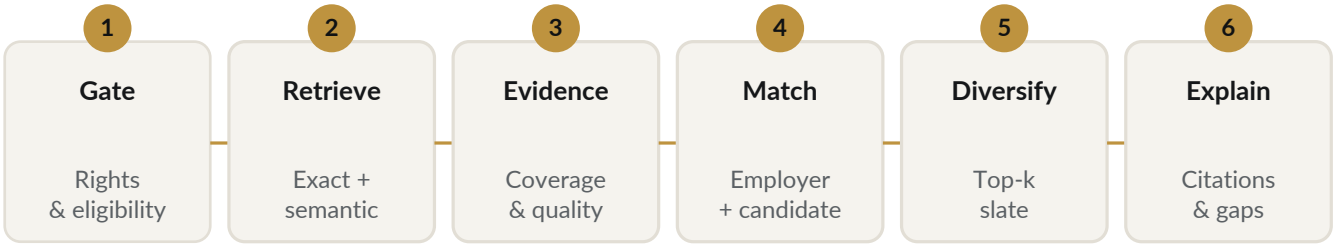
J role outcomes; H approved hard constraints; K competencies; W weights; P employment preferences; T assessments; G granted actions; V rubric version.

## Mandate quality gate

Reject ambiguous requirements such as “smart person with good culture fit” until they are converted to observable tasks and legitimate work conditions. Flag unrealistic conjunctions (e.g., rare skill bundle with low budget); offer trade-offs. A rubric can be updated, but a new version must trigger explicit reranking and preserve earlier results.

06 / RECOMMENDATION ARCHITECTURE

# Q-Match is a pipeline, not a magic score




## Stage 1: safe retrieval

Apply data entitlements and legitimate hard filters first. Retrieve candidate IDs via structured predicates (e.g., geography, certification where mandatory) plus lexical and vector search. Hybrid retrieval can use reciprocal-rank fusion:

$$RRF(i) = \Sigma_s [1 / (k + rank_s(i))]$$

s indexes retrieval systems; k is a positive smoothing constant chosen on retrieval benchmarks (e.g., tested values around 60).

RRF is only a candidate-generation mechanism. It does not assert a candidate is qualified. Scores from different sources must not be treated as comparable without normalization and testing.

### Stage 2: evidence-based reranking

- Build feature vectors from permitted competency evidence and preferences.
- Use deterministic scoring formulas initially; compare to validated alternative models later.
- Record exclusion reason, unresolved requirements, evidence coverage and candidate-interest state.
- Generate explanations only from cited facts and computed features; never free-form infer hidden traits.

> **Product contract**
> The UI should show **best documented match among the searched and permitted pool**, not **best person in Lagos**. Source coverage is necessarily incomplete.

07 / CORE MATHEMATICS

## Employer-side competency fit

Every role has n job-relevant competency dimensions. The weights w_j are non-negative and sum to 1. Candidate demonstrations are evaluated against a role-specific rubric, not compared against personal characteristics.

$$\Sigma(j=1..n)\ w_j = 1;\ 0 \le s_{ij}, c_{ij}, \mu_j \le 1$$

s_ij = demonstrated competency level; c_ij = evidence confidence; μ_j = prespecified neutral missing-evidence baseline.

$$F_i = 100 \times \Sigma_j\ w_j\ [c_{ij}\ s_{ij} + (1-c_{ij})\ \mu_j]$$

F_i = provisional employer-fit index, 0–100. This is an interpretable shrinkage heuristic, not a probability of successful performance.

$$C_i = \Sigma_j\ w_j\ c_{ij};\ U_i = 1 - C_i$$

C_i = weighted evidence coverage; U_i = unresolved evidence fraction.

## Why two separate numbers matter

A candidate with an attractive F score but low C has an uncertain evaluation. The UI must present both. We do **not** multiply confidence by fit and then portray missing data as poor performance. The baseline μ is a provisional method that must be tuned and audited for differential effects.

## Role weight construction

Start with job analysis: pairwise importance comparisons or a structured weighting workshop with an employer and a subject-matter expert. Cap any one dimension from dominating without documented justification. Version the rubric and compare inter-rater agreement before release.

### Example competency rubric

| Competency | Weight | Evidence source |
|---|---|---|
| Investigate alerts | 0.40 | Scenario with investigation notes |
| Query security logs | 0.30 | Practical log task |
| Communicate incident findings | 0.20 | Structured report rubric |
| Document containment decisions | 0.10 | Incident discussion |

This example is a design illustration only. Weights and competency definitions are not assumed valid across all SOC employers.

08 / TWO-SIDED MATCHING

# Candidate compatibility and mutual fit

The employer fit F asks whether evidence supports the role requirements. Candidate compatibility A asks whether the opportunity fits the candidate's own stated requirements. Neither can replace explicit interest.

$$A_i = 100 \times [\Sigma_k v_k a_{ik} / \Sigma_k v_k]$$

$a_{ik} \in [0,1]$ for **known** candidate preference k. Unknowns must be shown as unknown and omitted from the numerator and denominator only with an explicit low-coverage warning.

Compatibility dimensions can include candidate-stated salary expectations, location, permitted work format, desired schedule and career direction. Separate exact constraints from negotiable preferences.

$$M_i = \sqrt{(F_i \times A_i)}$$

The geometric mean penalizes one-sided matches; it is a heuristic index, not the probability a candidate will join or succeed.

## Crucial distinction: salary fit versus protected attributes

Compensation compatibility is job-related and candidate-controlled. Age, sex, ethnicity, religion, pregnancy, disability or proxies for protected characteristics are not inputs to routine Q-Match ranking. If a lawful occupational exception exists, it requires jurisdiction-specific review outside the ordinary scoring pipeline.

## Interest is a separate state

| Candidate state | Display language | Next action |
|---|---|---|
| Unknown | Potential match; interest not confirmed | Request permission-aware interest check |
| Interested | Confirmed interest at timestamp | Invite to next legitimate step |
| Not interested | Declined or opted out | Stop outreach and follow retention rules |
| Unavailable | Not currently available | Respect preferred future-contact settings |

09 / CALCULATIONS AND INTERPRETATION

# Worked example: three candidates

Synthetic SOC analyst mandate: w = (0.40, 0.30, 0.20, 0.10). Baseline μ_j = 0.50 for each dimension. Numbers are illustrative and not performance predictions.

| Candidate | Demonstrated s vector | Confidence c vector |
|---|---|---|
| A | 0.85, 0.80, 0.80, 0.70 | 0.95, 0.85, 0.90, 0.70 |
| B | 0.95, 0.90, 0.70, 0.85 | 0.40, 0.35, 0.55, 0.30 |
| C | 0.75, 0.85, 0.80, 0.80 | 0.95, 0.95, 0.90, 0.90 |

| Candidate | Employer fit F | Coverage C | Candidate A | Mutual M |
|---|---|---|---|---|
| A | 77.8 | 88.5% | 92 | 84.6 |
| B | 64.6 | 40.5% | 65 | 64.8 |
| C | 77.6 | 93.5% | 78 | 77.8 |

**Read the scores correctly**
Candidate B has ambitious claimed scores but sparse supporting evidence. Candidate A has stronger mutual compatibility. Candidate C has the best coverage. None should be automatically hired or rejected based on this table. The shortlist should highlight **missing verification work** and invite human comparison of evidence.

### What happens when salary is unknown?

Do not fill it from social media or past salary. Mark A as provisional and ask the candidate privately. If a recruiter insists on a definitive ranking, Q should warn that the employer-side rank is available but the mutual-fit rank is incomplete.

10 / SLATE DESIGN

# Shortlist optimization: not just top 15 scores

Naively selecting the 15 highest-index candidates can create a redundant shortlist of people with nearly identical backgrounds. A better objective covers relevant capabilities and distinct approaches without manipulating protected demographics.

$$S^* = \operatorname{argmax}_{|S| \le K} \left[ \sum_{(i \in S)} M_i + \lambda D(S) - \eta R(S) \right]$$

D = job-relevant coverage of distinct competencies/experience; R = redundancy among near-duplicate profiles; λ and η chosen by experiments.

Use a transparent greedy maximal marginal relevance approximation for an initial implementation. Re-rank with evidence and policy checks after every new mandate version or material evidence update.

## Minimum operational controls

- Do not let diversity optimization import protected attributes into individual scores.
- Allow a reviewer to inspect exactly which job-relevant dimension each candidate adds.
- Use consistent score versions within any shortlist comparison.
- Respect candidate visibility and representation rights across agencies.
- Never inflate an empty candidate pool into the requested K results.

## The stopping condition

The system can return fewer than K candidates. Its explanation should identify whether the limiting factor was legal constraints, missing sources, poor evidence, compensation mismatch or confirmed unwillingness to engage. These are very different outcomes.

11 / MODEL OUTPUT CONTRACT

# Uncertainty, explanation and human review

Every recommendation should be explainable as a chain of evidence. The language model renders the explanation; deterministic code controls the facts and the calculated values.

| Output field | Example | Control |
|---|---|---|
| Summary | "Strong log-investigation evidence" | Bound to an observed work sample |
| Supporting evidence | Assessment A-104, timestamp, rubric v3 | Clickable provenance |
| Unverified item | "Employer dates not verified" | Explicit uncertainty |
| Missing requirement | "No evidence of cloud detection" | No negative personal inference |
| Compatibility | "Hybrid arrangement confirmed" | Candidate-confirmed timestamp |
| Next action | "Invite for specialist review" | No automatic rejection |

## Human-in-the-loop is not a checkbox

Reviewers need adequate evidence, time, authority and the ability to override recommendations. A nominal approval click on a concealed AI decision is not meaningful oversight. UK privacy guidance and Dutch regulator guidance discuss the importance of meaningful human involvement [S5, S6].

## Counterfactual tests

For a candidate matched to a job, ask which *job-relevant evidence* would legitimately change the assessment. Example: obtaining an evaluated incident-response work sample could address a skill evidence gap. Never advise a candidate to change protected or irrelevant personal characteristics.

> **Trust rule**
> Separate three things: **facts** (source-backed), **judgments** (rubric-backed and contestable) and **forecasts** (calibrated only after sufficient data). The UI must not blur them.

12 / STATISTICAL LEARNING PLAN

# From heuristics to validated prediction

Hand-designed Q-Match indices establish a testable baseline, not predictive authority. A later model can learn probabilities of specific outcomes when enough representative, lawfully processed data exists.

> $p_t(i,j) = P(Y_t = 1 \mid X_{ij}, \text{segment}, \text{time})$
>
> Possible t: qualified at review, interview accepted, offer accepted. Keep separate outcomes rather than collapsing them into a vague "hireability" number.

## Data requirements

- A pre-registered outcome definition and clear time window for each model.
- Training labels from observed events, with reasons for missing outcomes and right-censoring.
- Recruitment-market and job-family stratification; no assumed transportability between Nigeria, UK and US.
- Leakage prevention: no post-interview events as features when predicting pre-interview suitability.
- Temporal holdout splits, reviewer-blind datasets and prospective validation before use in consequential decisions.

## Evaluation metrics

| Measure | Technical meaning | Why it matters |
|---|---|---|
| Precision@K | Relevant candidates among reviewed top K | Recruiter time saved |
| NDCG@K | Quality of ranked results vs expert relevance labels | Ranking effectiveness |
| Brier score / ECE | Probability calibration if predictive model exists | Honest confidence |
| Recall by subgroup | Qualified candidates retrieved across assessed groups | Discoverability risk |
| Override and appeals | Human corrections and substantiated objections | Accountability |
| Time to shortlist | Clock to first useful, confirmed shortlist | User value |

A recent 2026 review challenges overconfident validity claims drawn from weak study designs. TalentQ therefore needs its own prospective operational validation, not borrowed accuracy percentages [S11].

13 / RESPONSIBLE RECRUITMENT

# Bias testing and lawful selection

A global recruitment product must treat employment ranking as a sensitive, regulated use case from day one. The EU AI Act includes automated recruitment matching and ranking within high-risk employment examples [S2]; its current stand-alone high-risk obligations timetable extends to December 2027 [S3].

## Required audit layers

| Audit | What to test | Response |
|---|---|---|
| Feature review | Direct and proxy use of protected characteristics | Remove or justify lawfully, version policy |
| Access parity | Who can complete identity, interview and skill steps | Provide accessible equivalents |
| Selection-rate analysis | Differential screening progression by legally assessed groups | Investigate causes and remediation |
| Calibration | Are model probabilities reliable across subgroups? | Retrain, recalibrate or withdraw |
| Adversarial testing | Accent, formatting, name swaps, prompt injection | Test invariance / harden inputs |
| Reviewer quality | Rubric compliance and inter-rater agreement | Training and quality assurance |

## Jurisdictional examples

- **European Union:** AI Act high-risk recruitment considerations, privacy, transparency, documentation and oversight [S2, S3].
- **United Kingdom:** UK GDPR protections around significantly impactful solely automated decisions [S5].
- **New York City:** Local Law 144 bias-audit and notice rules apply to covered automated employment decision tools [S7].
- **Nigeria:** Section 37 of the Nigeria Data Protection Act restricts certain solely automated significant decisions [S8].

Country entry requires legal review of the actual deployed tool, employer responsibilities, hiring discrimination rules, retention, cross-border transfers, use of biometric/video information and candidate rights.

14 / EVALUATION ORCHESTRATION

# Q interviews and specialist assessments

Tavus-class video avatars, low-latency speech and natural-language interviewing are interfaces, not substitutes for assessment science. Candidate outcomes should rest on observable job-related answers and work products.

## Four evaluation modes

| Mode | Appropriate role | Infrastructure |
|---|---|---|
| Structured text or voice | Sales, service, administration | Question bank + rubric + transcript |
| Live conversational scenario | Business development, support | Realistic customer/problem roleplay |
| Sandbox work sample | Engineering, operations, cybersecurity | Constrained IDE/log/SQL/document tasks |
| Specialist provider assessment | Advanced cyber, regulated roles | Secure third-party integration and review |

## Interview protocol

- Notify the candidate of AI participation, processing purposes, recordings and available alternative arrangements.
- Anchor questions to the approved role rubric. Ask follow-ups that clarify evidence, not protected traits.
- Timestamp transcript turns and assessment artifacts; separate raw observation from evaluator judgment.
- Use human evaluation where required; provide recourse for technical issues and accommodation needs.
- Maintain candidate coaching/rehearsal data separately from employer hiring evidence.

**Cybersecurity example**
A Level-1 SOC applicant receives synthetic sign-in logs and a mock alert. The candidate explains triage and writes an incident note. Q records time-stamped steps and maps work to an approved rubric. A qualified analyst reviews the evidence. TalentQ does **not** claim to replicate a complete SIEM or outperform specialist cyber ranges.

15 / AUTONOMY AND PERMISSIONS

# Governed recruitment agents

Persistent agents should work across sessions, but their autonomy must be delegated explicitly. Agent output is auditable and scoped to a single organization, mandate and candidate permissions.

| Action | Default policy | Required safeguard |
|---|---|---|
| Search authorized sources | Automatic after mandate approval | Source entitlement, rate budget |
| Update evidence / match | Automatic within permission scope | Lineage, version, re-score audit |
| Draft candidate outreach | Automatic | Policy and contact preference checks |
| Send outreach | Approval or explicit standing delegation | Opt-out, frequency cap, anti-spam |
| Book interview | Candidate agreement + scheduling scope | Calendar availability and confirmation |
| Reject, advance, offer | Human decision | Recorded reviewer and rationale |

## Agent execution protocol

> Intent → Plan → Permission check → Idempotent tool call → Verify → Record → Report
>
> For long-running work, store checkpoints; retries must not duplicate invitations, charges or candidate state transitions.

- Require idempotency keys and outbox/event-delivery guarantees.
- Set cost and token budgets per mandate; on exhaustion, request authorization.
- Tool outputs and imported web text are untrusted; strip prompt instructions from source content.
- Disable unilateral agent policy changes or credential access.
- Provide a clear "stop all activity" control and replayable audit trail.

16 / SUPER-SPEED WITHOUT FICTION

# Latency architecture and performance budgets

The goal is immediate feedback and fast useful information, not a promise that all external data can be acquired instantaneously. The system should progressively reveal authorized and trustworthy results.

| User-visible event | Proposed target | Path |
|---|---|---|
| UI response | p95 < 300 ms | Optimistic UI + local state |
| First streamed text | p95 < 1.5 s | Small planner, context prefetch, stream |
| Re-rank indexed shortlist | p95 < 2 s | Cached features + deterministic scoring |
| First useful new-search set | p95 < 5 s where sources allow | Parallel source dispatch + progressive results |
| External verification complete | No universal instant SLA | Async connector and visible status |
| Real-time voice turn | Provider-specific measured target | Streaming audio + interruption handling |

## Performance design

- Precompute embeddings and candidate competency vectors; update incrementally with change events.
- Route exact structured filters to indexes and use vector retrieval only when semantics add value.
- Separate retrieval, policy checks, scoring and explanation. Parallelize independent steps.
- Cache by tenant, mandate version, source-entitlement hash and data version.
- Avoid serial voice → LLM → search → LLM chains on simple tasks; speculative work must remain permission-safe.

```
T_useful = max(T_index, T_remote-first) + T_rank + T_render
```

An illustrative critical-path model. Total completion time includes slow connectors; the first useful result should not wait on every connector.

Measure p50/p95/p99 by jurisdiction, tenant size, data freshness, provider, query class and concurrent load. Collect results on realistic networks, including low-bandwidth mobile usage.

17 / ENGINEERING SPECIFICATION

# Storage, API boundaries and event contracts

## Minimum relational core

| Table | Primary linkage / concern |
|---|---|
| organizations / tenants | Multi-tenant isolation and region controls |
| users / memberships | Roles, inviter, expiry, account recovery |
| candidates / candidate_consents | Candidate identity and data rights |
| mandates / mandate_versions | Approved requirements and history |
| competencies / ontology_versions | Versioned role skill graph |
| claims / evidence / verifications | Source provenance and verification lifecycle |
| matches / match_explanations | Inputs, indexes, rubric and model versions |
| applications / interviews / assessments | Candidate events, outcomes and human review |
| agent_jobs / action_log / outbox | Delegation, retries and operational trace |
| deletion_requests / audit_events | Access history, correction and erasure |

## Representative endpoint contracts

POST /v1/mandates/{id}/compile → rubric_draft + gaps + version

Requires recruiter role; employer approval before search.

POST /v1/mandates/{id}/matches → job_id + snapshot_id

Returns an asynchronous job reference, stream updates and idempotency key.

GET /v1/matches/{snapshot_id} → list + F/A/M/C + evidence_refs

Response never expands evidence outside the current grant.

POST /v1/agents/{job_id}/approve-action → action_receipt

Requires specific approval scope; append audit event.

Public APIs require authentication, authorization and tenant-scoped policy evaluation per request. Use provider adapters so interview, email, ATS and model vendors can be substituted independently.

18 / NON-NEGOTIABLE CONTROLS

# Security, privacy and data lifecycle

## Threats that matter specifically to TalentQ

| Threat | Why it matters | Control |
|---|---|---|
| Cross-tenant data leak | Candidate history crosses employer boundaries | Tenant isolation, ABAC/RLS, adversarial tests |
| Prompt injection | Malicious CV/web text instructs agent to act | Treat retrieved content as data; tool policy gates |
| Privilege escalation | Agent sends unauthorized messages | Short-lived scopes, approval, action logs |
| Candidate impersonation | Fraudulent representation or fabricated credentials | Risk-based verification and disputes |
| Sensitive data exposure | Recordings and employment history are high-impact | Encryption, minimization, lifecycle deletion |
| Provider outage/cost spike | Third-party voice/sourcing tools unavailable | Circuit breakers, quotas, fallback paths |

## Data-rights semantics

- Candidate export, correction, withdrawal, access logs and deletion requests must reach linked caches, vector indexes and downstream providers according to retention obligations.
- Explicitly separate consent to store a CV from permission to share a profile with a particular recruiter.
- Bound recordings to stated assessment purpose; implement short default retention subject to law and employer obligations.
- Protect assessor notes from cross-employer reuse; reuse candidate work samples only when permitted.
- Apply scoped encryption keys, secret rotation, documented incident response and independent security review.

**Build gate**
No production outreach or AI-assisted ranking for consequential decisions before a documented data-protection assessment, fairness evaluation plan, access-control penetration tests and jurisdiction-specific policy approval.

19 / WHAT HAPPENS WHEN THINGS FAIL

# Operational resilience and failure recovery

| Failure | Incorrect behavior to avoid | Required recovery |
|---|---|---|
| No search results | "There is nobody suitable" | State which sources ran; offer calibrated changes |
| LinkedIn integration unavailable | Background scraping workaround | Explicit unsupported-source notice |
| Candidate profile conflict | Silent overwrite | Keep source versions + request verification |
| Voice call disconnect | Restart interview from zero | Resume consented session/checkpoint |
| Ranking model timeout | Invent LLM ranking | Show cached snapshot as dated or fail safely |
| Applicant opts out | Continued follow-ups | Stop agent actions and process request |
| Assessment provider outage | Mark candidate failed | Pause, reassign or accessible alternative |
| Candidate challenges assessment | Auto-dismiss appeal | Qualified human review with trace |

## Operational event examples

**mandate.approved**, **source.sync_failed**, **evidence.verified**, **candidate.opted_out**, **match.recomputed**, **interview.disconnected**, **assessment.reviewed**, **application.human_decision_recorded**. Each carries tenant ID, entity ID, occurred_at, actor, schema_version and idempotency key.

20 / PROVE BEFORE CLAIMING

# Evaluation suite and acceptance gates

| Gate | Benchmark / artifact | Go/no-go principle |
|---|---|---|
| G0 Data rights | Tenant and source-entitlement tests; deletion drill | Zero known authorization bypasses |
| G1 Mandate accuracy | Expert review of structured role extraction | Clarification before uncertain commitments |
| G2 Search relevance | Labeled role-specific dataset; Precision@K/NDCG | Beat simple keyword baseline |
| G3 Fit quality | Blinded expert assessment; disagreement analysis | Credible, reproducible explanations |
| G4 Fairness & accessibility | Subgroup analyses, accommodations, audit docs | No unresolved material harm |
| G5 User outcomes | Matched control pilots and measured recruiter hours | Evidence of real customer improvement |
| G6 Speed & reliability | Load tests, p95/p99, retries and outages | Targets met on representative workloads |
| G7 Paid value | Renewal/repeat mandates and verified unit costs | Sustainable paid usage |

## Experiment design

Randomize or alternate appropriately matched recruitment mandates between an existing recruiter workflow and TalentQ assistance where customers agree. Record recruitment hours, candidates reviewed, interviews accepted, employer rating of evidence, candidate drop-off and placement outcomes. Do not infer future on-job performance from a small pilot.

## Pre-registered stop conditions

- Discovery legally blocked by data licensing or retention conditions.
- Bias audit or accessibility findings cannot be mitigated promptly.
- Outreach complaint rate exceeds an agreed threshold.
- Actual per-mandate delivery cost overwhelms pricing in repeat pilots.

- Q-Match performs no better than cheaper keyword / rule-based alternatives.

21 / TECHNICAL ECONOMICS

# Costs, unit budgets and scale

The market research report will estimate real provider costs and competitive pricing. This engineering model defines what to instrument before proposing margins.

$$Cost(m) = \Sigma_p\ units(m,p) \times rate(p) + storage(m) + egress(m) + review(m)$$

p includes model tokens, voice minutes, video sessions, enrichment calls, assessment provider events and message delivery. Rates are contractual and geography-dependent.

$$Gross\ contribution(m) = Revenue(m) - direct\ variable\ cost(m)$$

Exclude sunk product R&D; include paid human review when it is needed to fulfil the mandate.

## Cost-control mechanisms

- Use smaller models for extraction and candidate-format normalization; reserve stronger reasoning for exceptional ambiguity and explanation review.
- Deduplicate enrichment events and avoid full reindexing on minor changes.
- Meter premium video interviews, verification and specialist assessment usage.
- Place per-tenant and per-mandate quotas on sourcing queries, outreach and models.
- Prefer cached evidence snapshots for repeat recruiter interactions, with visible freshness labels.
- Require manual authorization before unusually expensive agent plans.

## Scale architecture

A modular-monolith baseline reduces operational overhead. Split the ingestion worker, search cluster or interview media subsystem only when performance and risk telemetry demonstrates the need. Regional tenancy, data residency and processor-contract support should exist in the schema before international deployment.

22 / DELIVERY ORDER

# Phased engineering roadmap

| Phase | Build | Exit evidence |
|---|---|---|
| P0 / Design & assurance | Mandate taxonomy, privacy architecture, skill ontology, jurisdiction review | Approved data model and safety controls |
| P1 / Recruiter pilot | Mandate compiler, ATS import, source-limited search, deterministic Q-Match | Blind shortlist evaluation on real mandates |
| P2 / Candidate loop | Native profile, consent UI, interest checks, scheduling, status | Candidate completion and trust metrics |
| P3 / Evaluation layer | Structured Q interviews, simple work samples, human review | Assessment consistency and fairness testing |
| P4 / Agent operations | Mandate-scoped agents, outreach approvals, retry orchestration | Measured hours saved, controlled costs |
| P5 / Outcome models | Longitudinal data, calibration, monitoring, predictive trials | Prospective validation before deployment |

## Technical definition of done for every module

- An owner, product contract, access policy, validation metrics and budget.
- Threat modeling, unit/integration tests and representative failure recovery.
- Versioned schemas, source auditability and opt-out/delete semantics.
- Latency/cost instrumentation and externally observable error states.
- A tested replacement path for provider dependencies.

**Critical design judgment**
Do not build a full learning management system, custom SIEM, global applicant-tracking suite or proprietary voice engine in phase one. Connect the specialist capabilities needed by real recruitment journeys; build deeper only when external integrations fail commercially or technically.

23 / WHAT IS REAL VS PROPOSED

# Investor claim register

| Claim | Current classification | Proof needed |
|---|---|---|
| "Q-Match exists" | Design proposal in this blueprint | Implemented scoring service + reproducibility tests |
| "Q-Match predicts successful hires" | Not supported | Prospective outcomes and calibration |
| "All LinkedIn talent is searchable" | Not supported; access restricted | Executed partner agreement / lawful source contract |
| "Recruitment agents run end to end" | Proposed operational design | Permissioned integrations and safe completion logs |
| "TalentQ has a data moat" | Potential future moat | Scale of consented evidence and outcome signals |
| "Sub-second search" | Target only for indexed/known sets | p95 benchmark at representative load |
| "Assessments are fair" | Unproven until audited | Validity, accessibility and impact tests |

## Pitch-safe technical statement

> **Defensible wording**
> “TalentQ is developing an evidence-driven, two-sided matching system that links job requirements, candidate-verified capabilities and candidate preferences. Its planned architecture provides source-aware explanations and human-governed recruitment workflows, with predictive models subject to prospective validation.”

There is no need to disclose internal reuse of the Q runtime in the public pitch. If an investor asks about ownership, dependency or intellectual property, answer fully and accurately about the legal and technical relationships. Marketing independence is not a license to misrepresent IP ownership.

24 / UNRESOLVED QUESTIONS

# Open decisions for engineering and diligence

| Decision | Proposed baseline | Evidence required |
|---|---|---|
| First job family | B2B sales or technical support plus one specialist test | Employer demand, assessment practicality |
| Candidate source model | Native plus authorized employer ATS imports | Actual access rights and candidate availability |
| Candidate compensation privacy | Ranges with explicit consent; no historical salary inference | User interviews and jurisdiction review |
| Matching model | Rubric-driven transparent heuristic | Expert labels, bias tests, comparative benchmarks |
| Recruiter autonomy | Draft first, send only with authorization | Adoption and opt-out measurement |
| Interview mode | Text/voice first, video optional | Completion, cost and accommodations |
| Data geography | Jurisdiction-aware tenant partitions | Contractual and legal analysis |
| AI provider strategy | Adapter interfaces, no hard-coding | Latency, cost, safety evaluation |

## The next two companion artifacts

**Document 02: Global market and commercial diligence.** A research protocol with nested evidence tasks, competitor workflows, user complaints, regional TAM/SAM/SOM with reproducible assumptions, pricing, cost benchmarks, regulation, market entry and sensitivity scenarios. Source dates and confidence levels will be explicit.

**Document 03: Investor pitch deck.** Built only after comparing the technical feasibility and market findings. It should distinguish credible current assets, planned capabilities, market hypotheses and explicit investment scenarios; no invented traction, IP claims or investor-return guarantees.

APPENDIX A / PRIMARY SOURCES

# Research and regulatory references

**[S1] LinkedIn Talent Solutions / Recruiter System Connect technical overview**

https://learn.microsoft.com/en-us/linkedin/talent/recruiter-system-connect?view=li-lts-2026-04

**[S2] European Commission AI Act Service Desk / Employment high-risk use cases**

https://ai-act-service-desk.ec.europa.eu/en/employment-0

**[S3] European Commission / AI Act implementation timeline and 2026 amendments**

https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

**[S4] NIST / Artificial Intelligence Risk Management Framework 1.0**

https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10

**[S5] UK ICO / Rights related to automated decision-making and profiling**

https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/individual-rights/individual-rights/rights-related-to-automated-decision-making-including-profiling/

**[S6] Dutch Data Protection Authority / Automated assessment decisions**

https://autoriteitpersoonsgegevens.nl/en/themes/employment-and-benefits/job-applications/online-and-game-based-assessments-in-recruitment-and-selection-rules-for-automated-decision-making

**[S7] New York City DCWP / Automated Employment Decision Tools, Local Law 144**

https://www.nyc.gov/site/dca/about/automated-employment-decision-tools.page

**[S8] Nigeria Data Protection Act 2023, section 37 (public transcription)**

https://lawcarenigeria.com/nigeria-data-protection-act-2023/

**[S9] ISO/IEC 42001:2023 / AI management systems, IEC**

https://webstore.iec.ch/en/publication/90574

**[S10] 2026 selection-science review, Annual Review of Organizational Psychology**

https://www.annualreviews.org/content/journals/10.1146/annurev-orgpsych-020924-072127

**[S11] Niessen et al., 2026 / Critique of selection criterion validity estimates**

https://onlinelibrary.wiley.com/doi/10.1111/ijsa.70086

**[S12] NIST AI RMF / Generative AI Profile**

https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence

Accessed / checked 25 October 2026. Sources inform risk and technical context; they do not validate TalentQ performance. This is a design document, not legal or statistical certification.

APPENDIX B / DETERMINISTIC MATCH FLOW

# Algorithm reference pseudocode

| Step | Operation | Output / control |
|---|---|---|
| 01 | Authorize request | Check tenant, recruiter role, mandate approval, source grants and candidate visibility. |
| 02 | Compile constraints | Normalize job tasks, validated eligibility, rubric and candidate preference slots. |
| 03 | Generate pool | Structured search plus licensed lexical/semantic retrieval; deduplicate. |
| 04 | Resolve evidence | Fetch source-linked evidence within scope, freshness and contradiction status. |
| 05 | Score employer fit | For each role competency, compute observed score, confidence and prior baseline. |
| 06 | Compute mutual fit | Candidate preferences, compatibility coverage and geometric index; preserve unknowns. |
| 07 | Optimize shortlist | Greedy relevance/coverage optimization under legitimate constraints. |
| 08 | Explain, audit, return | Evidence-bound explanations, snapshot, versions, action and policy logs. |

## Version string carried with every match

```
match_version = hash(mandate_v, rubric_v, source_snapshot, policy_v, algorithm_v)
```

This allows deterministic reproduction, differential regression testing and audit of why rankings changed.